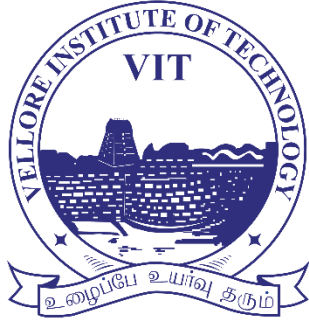




VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

DA – Case Study

Project: Adaptive Lamp with Distance-Based Dynamic Illumination Control

Name: A.RAMESH KUMARAN

Reg.No.: 22MIC0072

Course: Advanced Data Visualization Techniques

Course code: CSI3005

Submitted to : Dr. YUVARAJ N



Adaptive Lamp with Distance-Based Dynamic Illumination Control

1. Problem Description:

Poor reading distance and non-optimal lighting contribute to eye strain, especially for children and individuals who spend long hours reading or studying. Traditional lamps provide static illumination without awareness of the user's reading posture or distance. This project proposes an adaptive intelligent lamp that continuously monitors the user's distance from the reading surface and automatically adjusts brightness and color temperature to optimize visual comfort.

2. Key Features

- Real time distance measurement using an ultrasonic sensor or small camera.
- Automatic adjustment of brightness, color temperature, or both, based on distance thresholds.
- Visualization module showing distance trends, illumination levels, and eye strain risk indicators.
- Personalized learning model adapting illumination preferences based on user habits.
- Eye comfort feedback system that notifies users when they sit too close or too far.

3. Novelty Components

- A distance-adaptive illumination algorithm that dynamically balances brightness and color temperature.
- A closed loop feedback mechanism linking user distance, illumination changes, and comfort metrics.
- Novel method for predicting eye strain based on distance patterns and illumination history.
- Intelligent user-specific illumination profiles that evolve automatically through behavioural learning.
- Unique workflow that combines reading distance sensing with physiological comfort optimization.

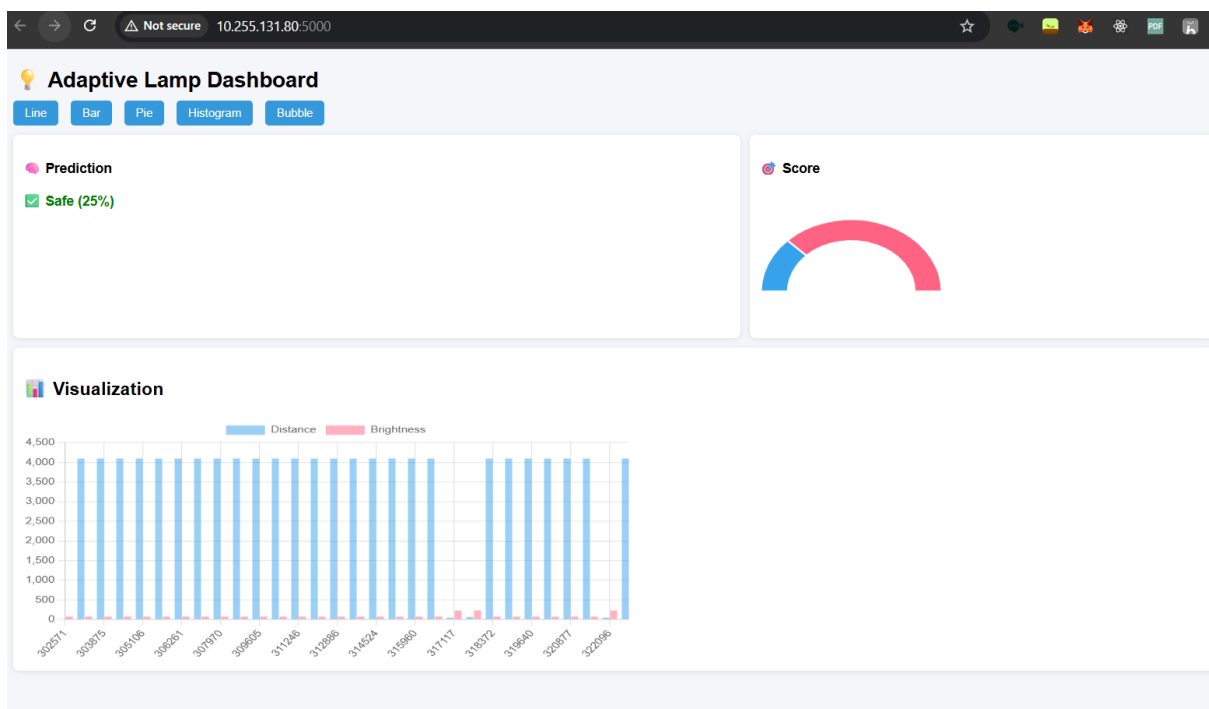
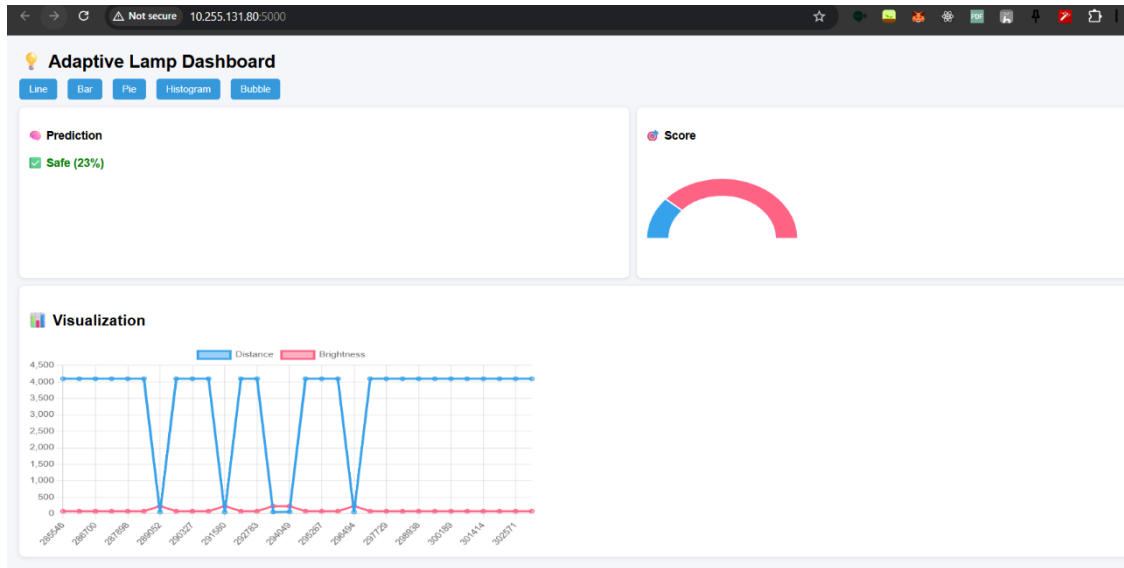
4. Expected Outcomes

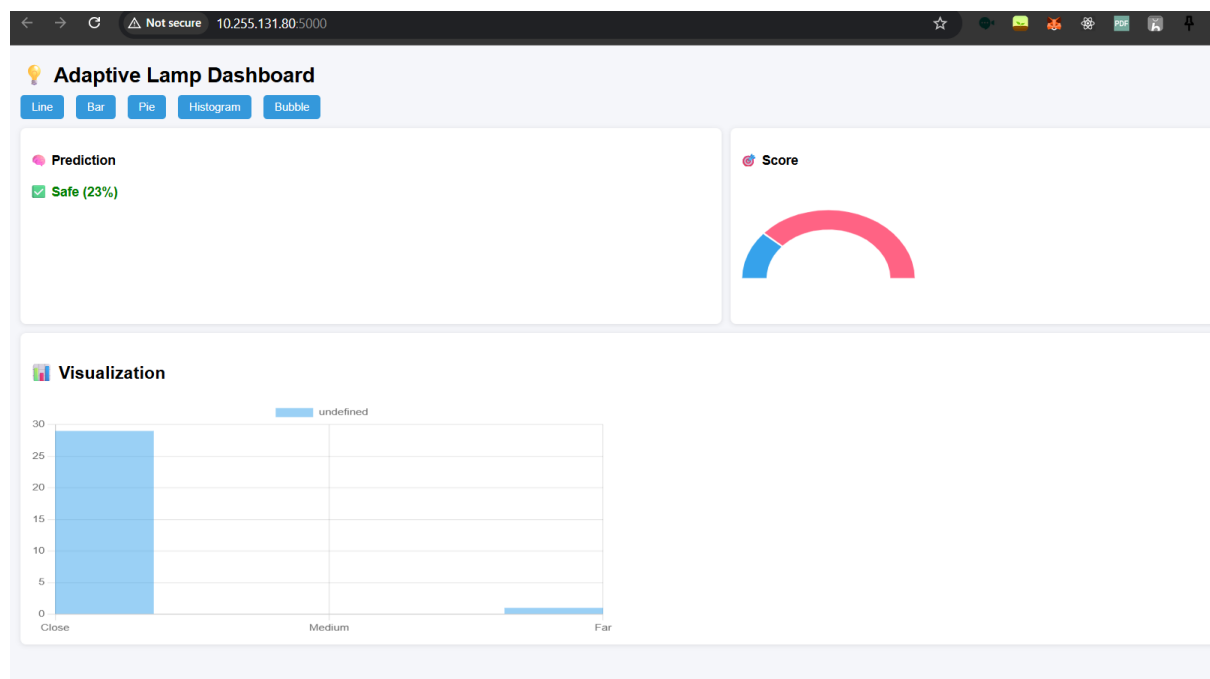
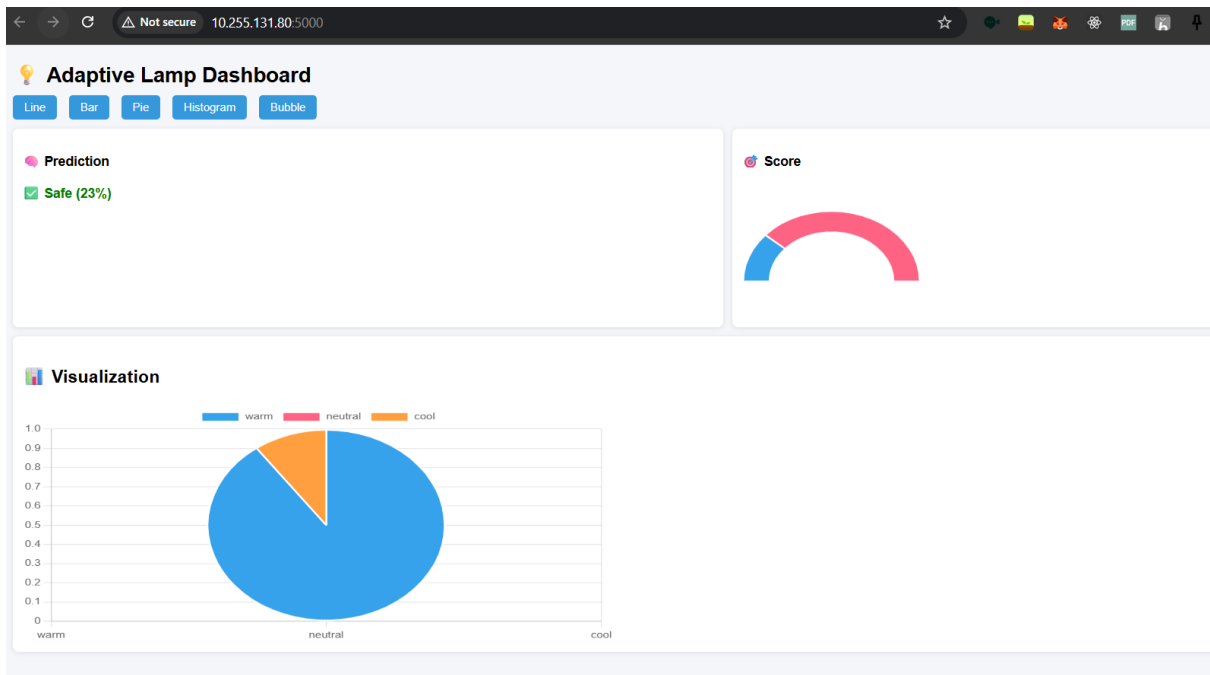
- Functional prototype of an adaptive reading lamp.
- Real time visualization of user distance and illumination adjustments.
- Predictive alerts for eye strain based on reading posture.
- Personalized lighting adaptation model.

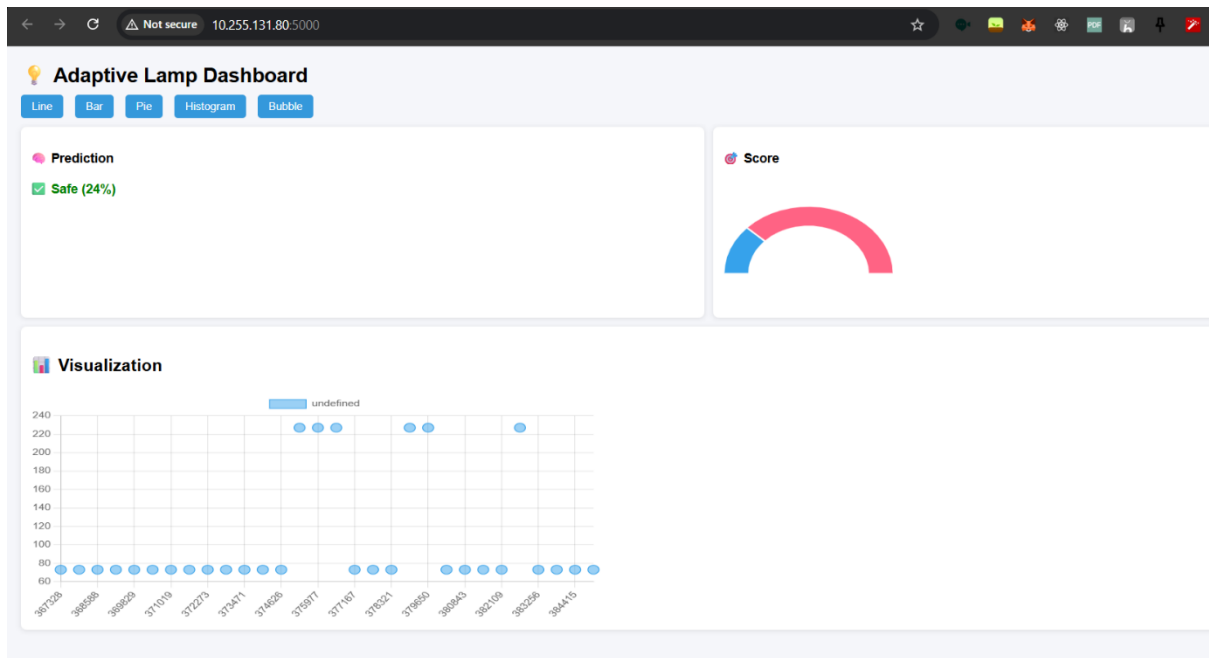
Github Link: <https://github.com/ARamesh-tech/Adaptive-Lamp-Light>

OUTPUTs:

All the visualizations are from the real time data which changes with time.







Performance metrics for the models:

➤ RandomForestClassifier and LSTM:

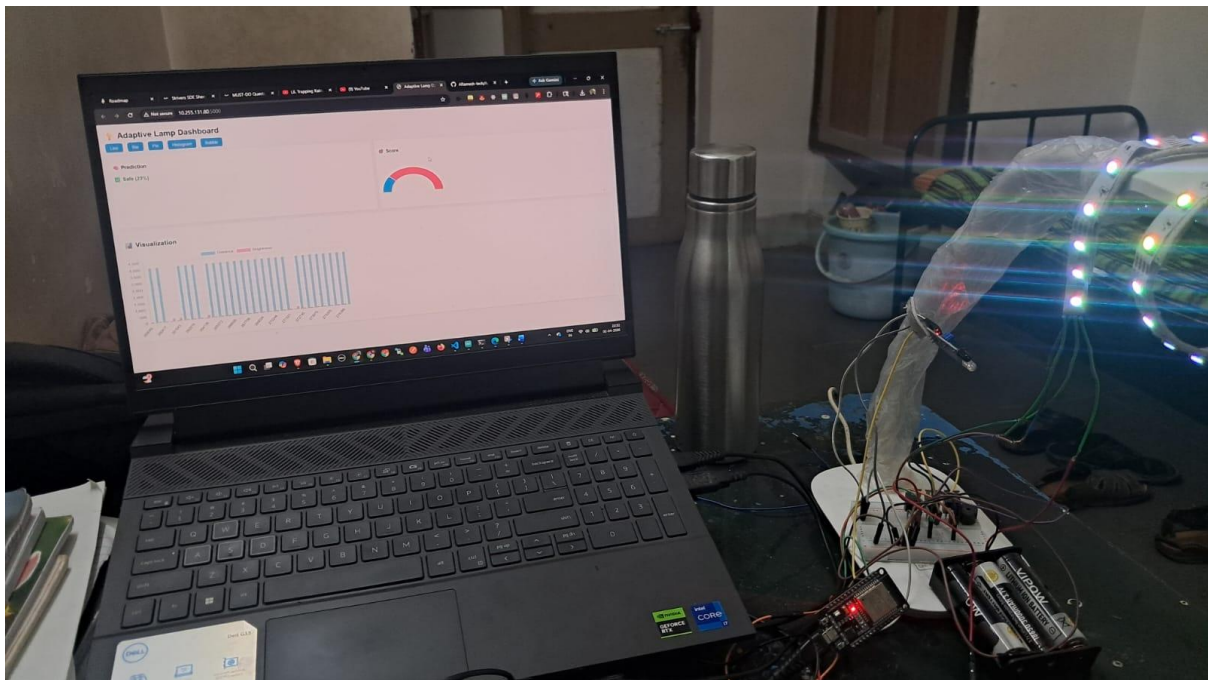
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell +
PS D:\Projects\AdaptiveLampLight> python '.\models\train_model.py'

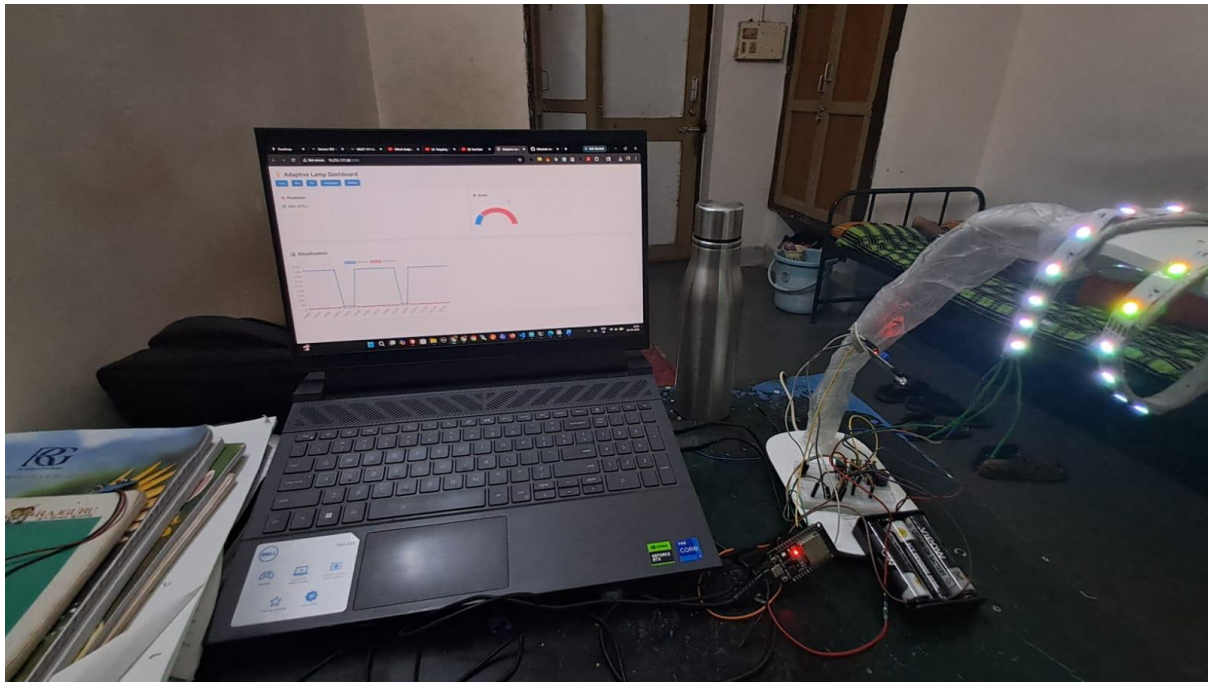
Class Distribution:
strain
0    997
1    166
Name: count, dtype: int64

Random Forest Metrics
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0
Epoch 1, Loss: 1.1859240531921387
Epoch 2, Loss: 1.1836646795272827
Epoch 3, Loss: 1.1814402341842651
Epoch 4, Loss: 1.1792480945587158
Epoch 5, Loss: 1.1770843267440796
Epoch 6, Loss: 1.174943447113037
Epoch 7, Loss: 1.1728196144104004
Epoch 8, Loss: 1.1707053184509277
```

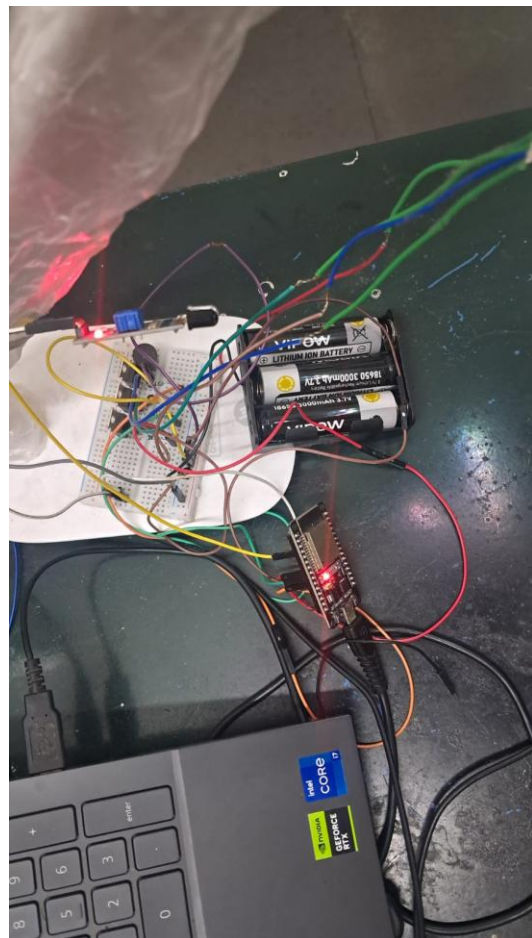
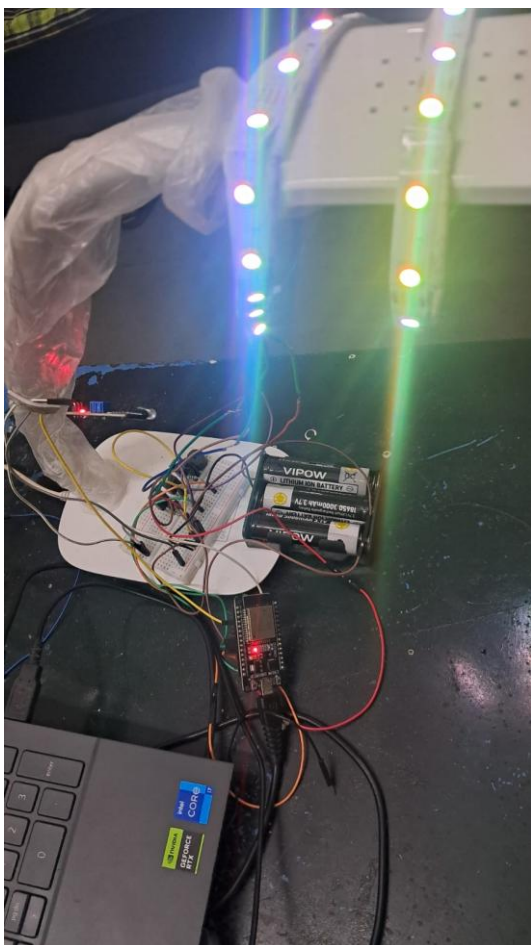
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Projects\AdaptiveLampLight> python '.\models\train_model.py'
Epoch 4, Loss: 1.1792480945587158
Epoch 5, Loss: 1.1770843267440796
Epoch 6, Loss: 1.174943447113037
Epoch 7, Loss: 1.1728196144104004
Epoch 8, Loss: 1.1707053184509277
Epoch 9, Loss: 1.1685930490493774
Epoch 10, Loss: 1.1664751768112183
Epoch 11, Loss: 1.164344072341919
Epoch 12, Loss: 1.1621919870376587
Epoch 13, Loss: 1.1600109338760376
Epoch 14, Loss: 1.1577941179275513
Epoch 15, Loss: 1.155534029006958
Epoch 16, Loss: 1.1532241106033325
Epoch 17, Loss: 1.1508573293685913
Epoch 18, Loss: 1.148427128791809
Epoch 19, Loss: 1.1459259986877441
Epoch 20, Loss: 1.1433473825454712
Epoch 21, Loss: 1.1406837701797485
Epoch 22, Loss: 1.1379274129867554
Epoch 23, Loss: 1.1350715160369873
Epoch 24, Loss: 1.1321076154708862
Epoch 25, Loss: 1.129028558731079
Epoch 26, Loss: 1.1258262395858765
Epoch 27, Loss: 1.1224939823150635
Epoch 28, Loss: 1.1190242767333984
Epoch 29, Loss: 1.1154102087020874
Epoch 30, Loss: 1.111646294593811

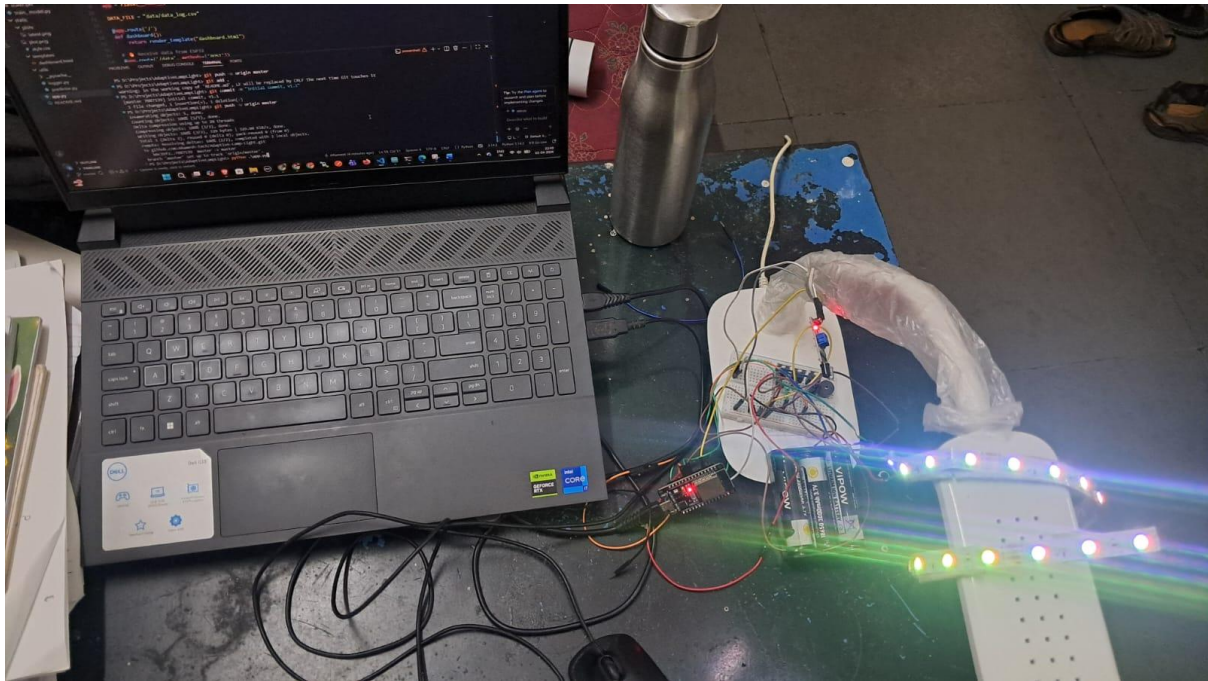
◆ LSTM (PyTorch) Metrics
Accuracy: 0.15517241379310345
Precision: 0.15517241379310345
Recall: 1.0
F1 Score: 0.26865671641791045
❖ PS D:\Projects\AdaptiveLampLight>
```



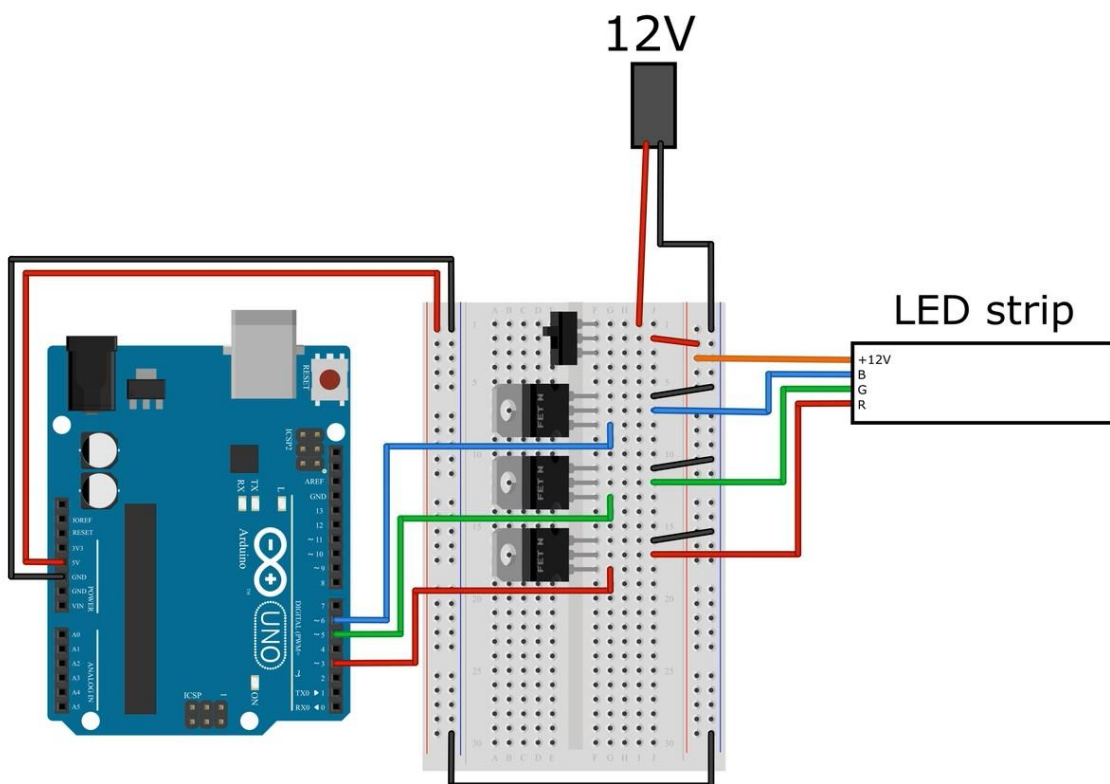


Circuit setup:

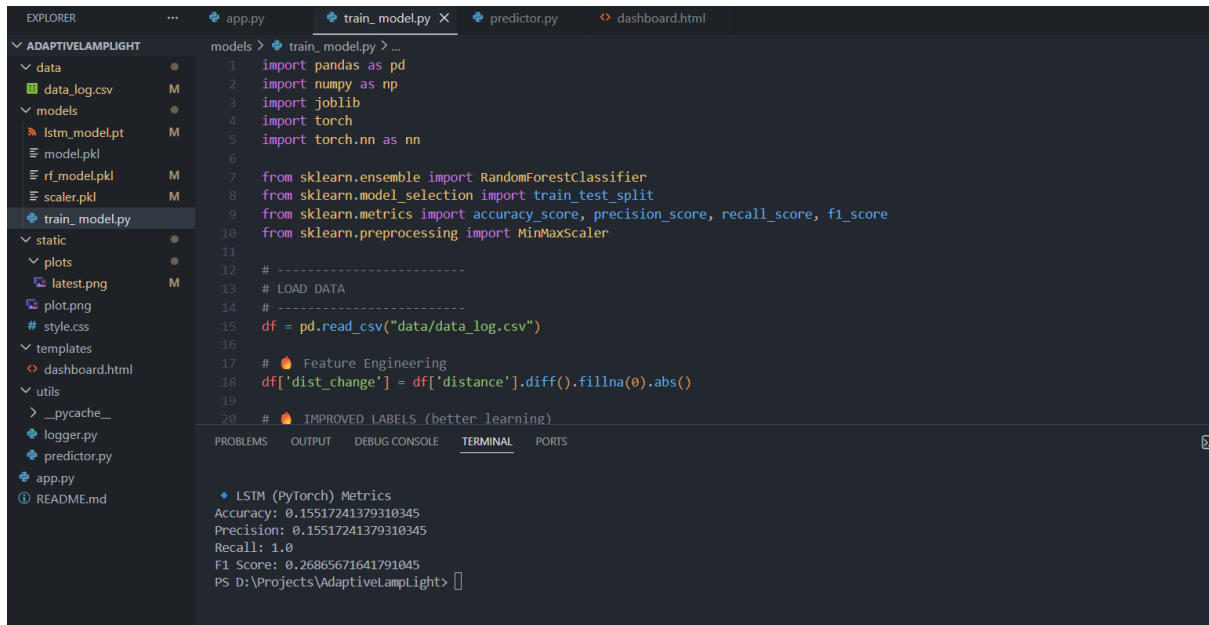




Circuit connections for RGB LED Strip to control Light brightness and color temperature: use either Arduino or ESP32



Project Directory:



The screenshot shows a VS Code editor interface. On the left, the Explorer sidebar displays the project directory structure for 'ADAPTIVELAMPLIGHT'. The files listed are: data (folder), data_log.csv (file), models (folder), lstm_model.pt (file), model.pkl (file), rf_model.pkl (file), scaler.pkl (file), train_model.py (file), static (folder), plots (folder), latest.png (file), plot.png (file), style.css (file), templates (folder), dashboard.html (file), utils (folder), __pycache__ (folder), logger.py (file), predictor.py (file), app.py (file), and README.md (file). The main editor area shows the 'train_model.py' file with the following code:

```
1 import pandas as pd
2 import numpy as np
3 import joblib
4 import torch
5 import torch.nn as nn
6
7 from sklearn.ensemble import RandomForestClassifier
8 from sklearn.model_selection import train_test_split
9 from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
10 from sklearn.preprocessing import MinMaxScaler
11
12 # -----
13 # LOAD DATA
14 # -----
15 df = pd.read_csv("data/data_log.csv")
16
17 # 🔥 Feature Engineering
18 df['dist_change'] = df['distance'].diff().fillna(0).abs()
19
20 # 🔥 IMPROVED LABELS (better learning)
```

At the bottom of the editor, the TERMINAL tab is active, displaying the following output:

```
♦ LSTM (PyTorch) Metrics
Accuracy: 0.15517241379310345
Precision: 0.15517241379310345
Recall: 1.0
F1 Score: 0.26865671641791045
PS D:\Projects\AdaptiveLampLight>
```

Visit Github Link to access the complete project (Adaptive Lamp Light):

<https://github.com/ARamesh-tech/Adaptive-Lamp-Light>

Submitted to:

Dr. Yuvraj N

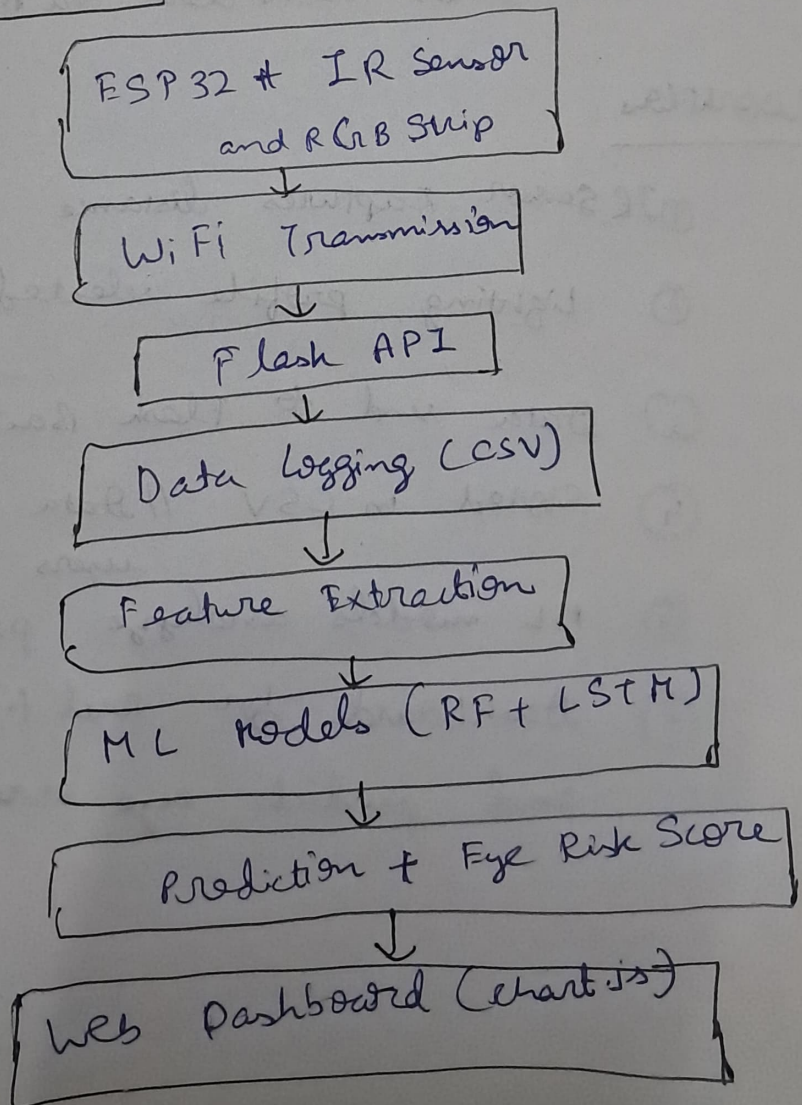
Name: A. Ramesh kumaran

Reg. No: 22MICO072

Objectives:

- To develop an adaptive lighting system based on user proximity
- Reduced eye strain using data-driven predictions
- Integrate IoT + ML + web Dashboard for real time monitoring
- Provide personalized lighting recommendations

System Architecture:



Hardware Setup:

- Components:

- ESP32 microcontroller

- IR Distance Sensor

- RGB LED

- Functionality:

- Reads analog distance values

- Maps distance $\Rightarrow$ Lighting profile
(^① Warm; ^② neutral;

- ^③ cool)
 - Sends JSON data via HTTP to Flask server

Workflow:

① IR sensor captures distance

② Lighting profile selected (color + brightness)

③ Data send to Flask Backend

④ Stored in CSV // Data collection for users.

⑤ ML models analyze patterns

⑥ Dashboard for real time visualizations and predicts eye strain.